



Intel 8th Gen core CPUs

Intel launches Kaby Lake refresh CPUs. Core i Series 8, but not Coffee Lake! Confuses everyone.
<http://dgit.in/CoffeeLakeU>



\$200 price cut for Vive

HTC is cutting down the price for its virtual reality headset by \$200 bringing its total cost down to \$599. Learn more about it here: <http://dgit.in/200vive>

and long tail. Based on its strong body and small skull, it was named *Sclerocormus parviceps*. The creature was alive when the Earth had just experienced the "Great Dying" event, or the End-Permian mass extinction, around 252 million years ago. Fresh from the catastrophe and after a few million years, many creatures were believed to have been claiming their dominance over the planet. Back in 2014, the fossil of apparently the first amphibian was discovered, named *Cartorhynchus lentacarpus*, which displayed similarities to *Sclerocormus* but also was smaller in size and had a shorter tail. The fossil is said to be around 250 million years old, suggesting that creatures were drastically evolving. This was mainly because of the huge void left after the mass extinction that killed off nearly 96% of the species on the earth. In the span of a few million years, a diverse pool of creatures were already present in the waters and on land, pointing towards rapid evolution on a global scale in the Triassic period.

Although this was the oldest and probably the first known event of rapid evolution yet, it's known that mass extinction encourages it. The Cretaceous-Paleogene extinction event wiped off all the dinosaurs from the face of the earth. Yet another void was left in the ecosystem that had to be filled in by other species, mostly land creatures. Among the creatures that profited from the death of dinosaurs were frogs, according to a study. Most of the dinosaur population were herbivores and in their absence the vegetation flourished. This allowed frogs to take over the forests and become arboreal (start living in trees). There was abundant food for the frogs and the threat from predators was solved by another shift in their reproductive cycle. Some frogs were able to skip the tadpole phase and directly grow into adult frogs. As a result, approximately 88% of the current frog species originated in this period.

This might not be a surprise to many but even humans have been evolving rapidly, especially in the last 40,000 years. Considering this pace, if we were

to evolve the same way when humans and chimpanzees first diverged (dating back to 6 million years ago), the genetic differences would have been 160 times greater. This evolution was also motivated by the discovery of agriculture some 10,000 years ago. The main driving factors for this rapid evolution in humans include environmental changes and the population explosion which multiplied the number of mutations generated.

The above-mentioned events describe evolution over several generations and even millions of years. Although it's a really long period, these particular events highlight those instances where the evolution was quicker than conventional time periods. These events have been determined through fossil records but in recent times, rapid evolution is being observed in our lifetime in a shorter time scale of a few decades and few generations.

CAUSES OF RAPID EVOLUTION

We are primarily to blame for most of the recent rapid evolution events across organisms. Several instances of forced adaptations have been motivated by our own actions. Be it encroachment of natural habitats or artificial selection of species, many of these have been caused by us. And with such hostility, the species have been forced to evolve in order to survive. But not all organisms have been successful in evolving to survive from the ongoing mass extinction. You read that right! The Holocene extinction or the sixth mass extinction has been underway for thousands of years now, ever since we started dominating the environment, and it gains momentum every year. Population explosion and the resources required to sustain this enormous population have pushed many plants and animals towards extinction.

There have been several instances where organisms have undergone rapid evolution directly because of human activities leading to environmental changes. The peppered moth is a popu-

lar example from during the industrial revolution. Air pollution caused by coal smoke covered and darkened the barks of the trees in England, resulting in light pepper moths standing out to predators. Dark-coloured peppered moths found themselves in a favourable position and were able to survive and proliferate. This is pure natural



White-bodied peppered moth almost camouflaged

selection at play. The numbers of light peppered moths dwindled as birds were able to easily spot them. Eventually, when the air became cleaner, the barks went to their natural colour. This switched the advantage back to the struggling population of the light moths, and now dark-coloured moths were being easily spotted by predators.

Climate has always played an important role in the evolution of species. With the changing landmass and human encroachment, animals and plants have been forced to move into different areas with different climates. The only way to survive is to adapt to the local climate. You would naturally think that such an adaptation would take millions of years of evolution but a recent study suggests that it can be possible in just a single winter storm. A certain species of green Anole lizards in the southern part of US evolved resistance to cold weather after a record-breaking cold storm in 2014. Originally, the team of researchers from the University of Illinois was trying to study the capability of the cold-blooded reptiles to survive in the cold north. After the extreme storm hit the region, they went back to the region to collect data from the reptiles. They found that the lizards had developed 14 genomic

Credit: Andy Phillips (<http://dgit.in/AndyPMoth>)